

2015

YEAR 12

TRIAL

EXAMINATION

Mathematics

Extension 2

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using blue or black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.

Total marks – 100

- Attempt Sections I and II.
- Section I is worth 10 marks.
- Answer Section I on the multiple choice answer sheet.
- Detach the multiple choice answer sheet from the back of the examination paper.
- Section II contains 6 questions worth 15 marks each.
- Answer each question in a new booklet.
- Label all sections clearly with your name/number and teacher.

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

- 1 Write $\frac{30}{1-2i}$ in the form $a+bi$.
- (A) $10-20i$ (B) $-10-20i$ (C) $6+12i$ (D) $6-12i$
- 2 If $2-i$ is a root of the polynomial $x^3-7x^2+17x-15=0$, then the other roots are:
- (A) $2-i, 3$ (B) $2+i, 3$ (C) $2+i, -3$ (D) $2-i, -3$
- 3 The eccentricity of the hyperbola $\frac{x^2}{3}-\frac{y^2}{4}=1$ is:
- (A) $\frac{5}{3}$ (B) $\frac{\sqrt{21}}{3}$ (C) $1+\frac{2}{\sqrt{3}}$ (D) $\sqrt{\frac{7}{3}}$
- 4 If the polynomial equation $P(x)=0$ has roots α , β and γ , then the polynomial equation $P(2x-1)=0$ has roots:
- (A) $\frac{\alpha}{2}+1, \frac{\beta}{2}+1, \frac{\gamma}{2}+1$ (B) $\frac{\alpha+1}{2}, \frac{\beta+1}{2}, \frac{\gamma+1}{2}$
- (C) $2\alpha-1, 2\beta-1, 2\gamma-1$ (D) $\alpha-\frac{1}{2}, \beta-\frac{1}{2}, \gamma-\frac{1}{2}$
- 5 The gradient of the tangent to the curve $3x^2-y^2=11$ at $(-2,1)$ is:
- (A) -6 (B) 6 (C) $-\frac{3}{2}$ (D) $\frac{3}{2}$

6 The equation of the conic whose distance from the point $(2,0)$ is half its distance to the line $x=8$ is given by:

(A) $\frac{x^2}{16} + \frac{y^2}{12} = 1$ (B) $\frac{x^2}{4} + \frac{y^2}{3} = 1$ (C) $\frac{x^2}{16} - \frac{y^2}{12} = 1$ (D) $\frac{(x+4)^2}{48} + \frac{y^2}{24} = 1$

7 Given the curve $y = f(x)$, the curve $y = f(|x|)$ is represented by:

- (A) A reflection of the curve $y = f(x)$ in the x -axis.
(B) A reflection of the curve $y = f(x)$ in the y -axis.
(C) A reflection of the curve $y = f(x)$ for $x \geq 0$ in the y -axis.
(D) A reflection of the curve $y = f(x)$ for $y \geq 0$ in the x -axis.

8 If $x+y=1$ and $x^2+y^2=2$, then the value of x^3+y^3 is:

(A) $1\frac{1}{2}$ (B) 2 (C) $2\frac{1}{2}$ (D) $3\frac{1}{2}$

9 Which of the following gives an expression for $\int \frac{dx}{\sqrt{7-6x-x^2}}$?

(A) $\sin^{-1}\left(\frac{x+3}{2}\right) + C$ (B) $\sin^{-1}\left(\frac{x+3}{4}\right) + C$
(C) $\sin^{-1}\left(\frac{x-3}{2}\right) + C$ (D) $\sin^{-1}\left(\frac{x-3}{4}\right) + C$

10 Which of the following statements is FALSE?

(A) $\int_{-3}^3 x \cos(x) dx = 0$ (B) $\int_{-3}^3 e^{-x^2} dx = 2 \int_0^3 e^{-x^2} dx$
(C) $\int_0^{\pi} \sin^8 x dx > \int_0^{\pi} \sin(8x) dx$ (D) $\int_0^1 x^8 dx < \int_0^1 x^9 dx$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) $z = -2 + 3i$ and $w = 3 - 2i$ are complex numbers.
- (i) Express $z + w$ in modulus–argument form. 2
- (ii) Express $z \bar{w}$ in the Cartesian form. 2
- (b) Sketch the region of the Argand diagram whose points satisfy:
- $$\left| z - \bar{z} \right| \leq 2 \text{ and } -\frac{\pi}{4} \leq \arg(z) \leq \frac{\pi}{4}. \quad 2$$
- (c) The cube roots of unity are the solutions to $z^3 = 1$
- (i) Find the three roots in modulus-argument form and sketch them on an Argand diagram.. 4
- (ii) Write the three roots in Cartesian form. 1
- (iii) If one of the complex roots is ω , show that $1 + \omega + \omega^2 = 0$. 2
- (iv) Evaluate $(1 + \omega)(1 + \omega^2)(1 + \omega^5)(1 + \omega^8)(1 + \omega^{11})$. 2

Question 12 (15 marks) Use a SEPARATE writing booklet.

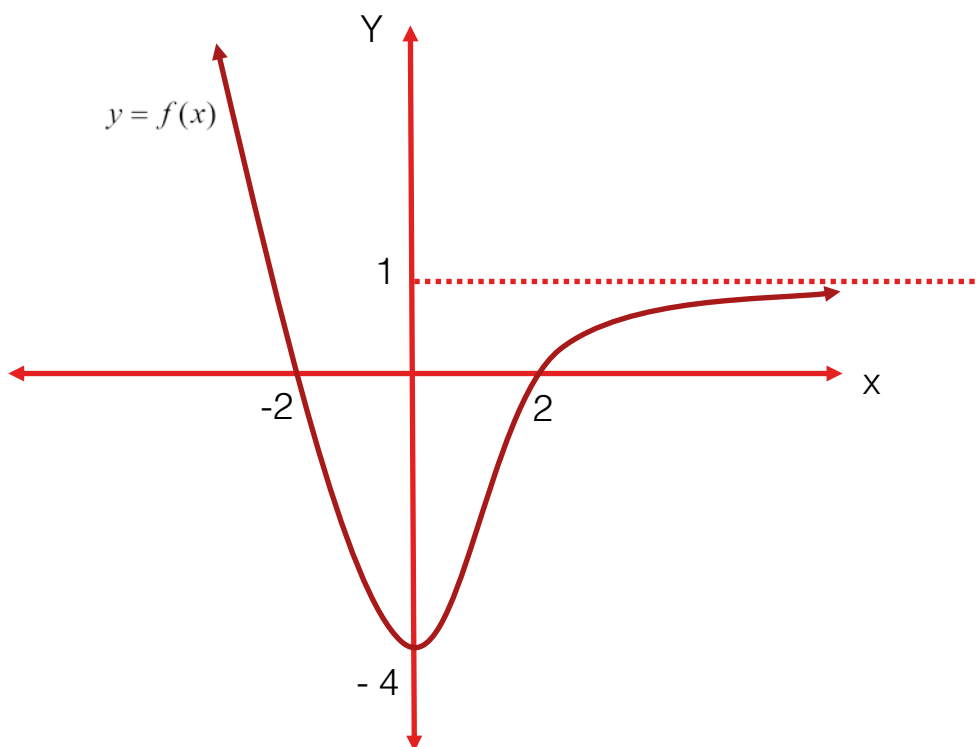
(a) Find $\int \cos^4 x dx$. 2

(b) Evaluate $\int_0^1 x e^{-x} dx$ 3

(c) (i) Express $\frac{x+7}{x^2(x+2)}$ in the form $\frac{A}{x^2} + \frac{B}{x} + \frac{C}{x+2}$. 2

(ii) Hence, or otherwise, find $\int \frac{x+7}{x^2(x+2)} dx$. 2

(d) The diagram below shows the graph of $y = f(x)$.



On separate half page graphs, sketch the following:

(i) $y = (f(x))^2$ 2

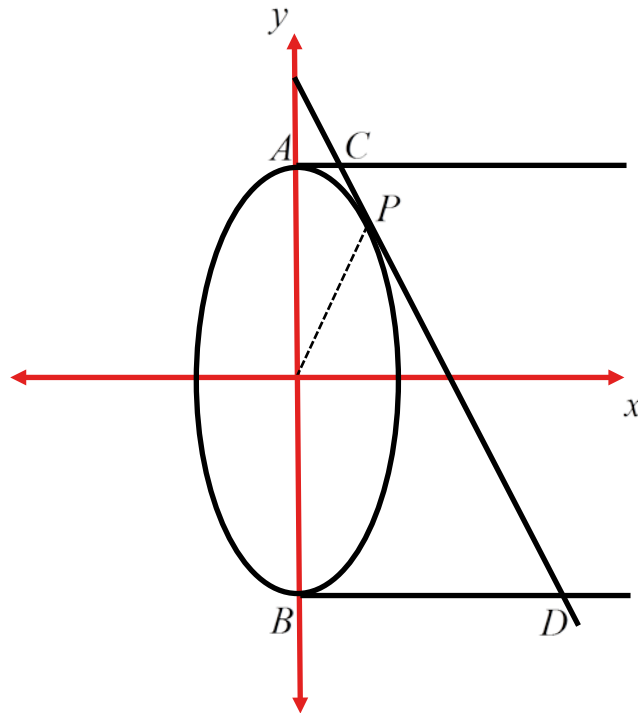
(ii) $y^2 = f(x)$ 2

(iii) $y = x \cdot f(x)$ 2

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram below shows the ellipse which has equation $\frac{x^2}{9} + \frac{y^2}{16} = 1$.

The point $P(3\cos\theta, 4\sin\theta)$ lies on the ellipse. The ellipse meets the y -axis at the points A and B . The tangents to the ellipse at A and B meet the tangent at P at the points C and D respectively.



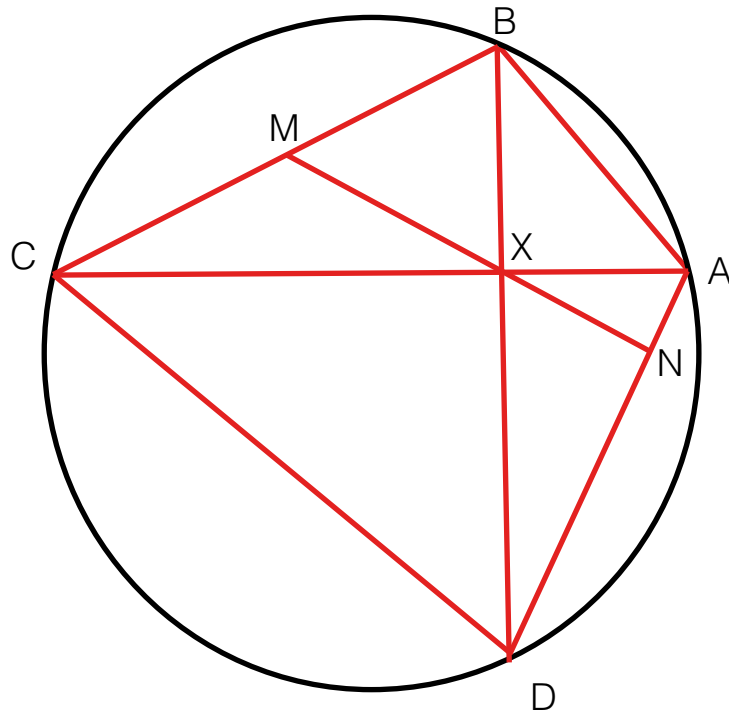
- | | | |
|-------|--|---|
| (i) | Find the eccentricity, coordinates of the foci and the equations of the directrices. | 4 |
| (ii) | Show that the equation of the tangent to the ellipse at P is $3y\sin\theta + 4x\cos\theta = 12$. | 2 |
| (iii) | Find the numerical value of $AC \times BD$. | 3 |
| (b) | (i) Sketch the region of the Cartesian plane bounded by:
$y = x^2$, $y = 1$, $x = 4$ and the X -axis. | 2 |
| | (ii) A volume is generated by rotating this region about the x -axis. Find this volume by using the method of slicing. | 4 |

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Show that the equation of the normal to the hyperbola $xy = c^2$ at $P\left(cp, \frac{c}{p}\right)$ is $p^3x - py = c(p^4 - 1)$. 2
- (ii) The normal at $P\left(cp, \frac{c}{p}\right)$ meets the hyperbola $xy = c^2$ again at $Q\left(cq, \frac{c}{q}\right)$. Prove that $p^3q = -1$. 2
- (iii) Hence, show that the locus of the midpoint R of PQ is given by $c^2(x^2 - y^2)^2 + 4x^3y^3 = 0$. 3
- (b) A particle of mass 1kg moves in a straight line against a resistance of $v + v^3$, where v is its velocity $\left(m\ddot{x} = -m(v + v^3)\right)$.
Initially the particle is at the origin and is travelling with velocity Q , where $Q > 0$.
- (i) Show that v is related to the displacement x by $x = \tan^{-1}\left(\frac{Q - v}{1 + Qv}\right)$. 2
- (ii) Show that the time t which has elapsed when the particle is travelling with velocity R is given by $t = \frac{1}{2} \log_e \left(\frac{Q^2(1 + R^2)}{R^2(1 + Q^2)} \right)$. 2
- (iii) Find R^2 as a function of t . 2
- (iv) Find the limiting values of v and x as $t \rightarrow \infty$. 2

Question 15 (15 marks) Use a SEPARATE writing booklet.

(a)



$ABCD$ is a cyclic quadrilateral. The diagonals AC and BD intersect at right angles at X . M is the midpoint of BC . MX produced meets AD at N .

(i) Copy the diagram showing the above information.

(ii) Prove that $\angle MBX = \angle MXB$. 3

(iii) Show that MN is perpendicular to AD . 4

(b) Find the equation with rational coefficients whose roots are the reciprocals of the squares of the roots of $x^3 - x^2 - 14x - 24 = 0$. 3

(c) (i) Show that $\sin x + \sin 3x = 2\sin 2x \cos x$. 2

(ii) Hence or otherwise find all solutions of: 3

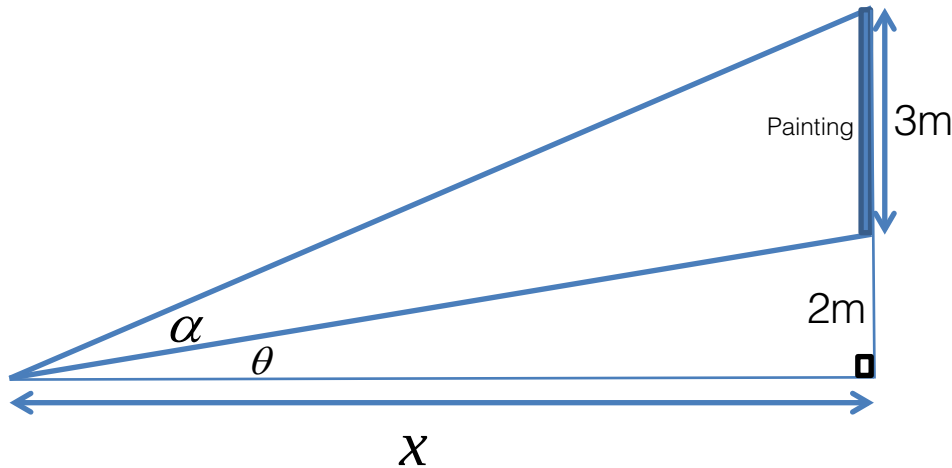
$$\sin x + \sin 2x + \sin 3x = 0 \text{ for } 0 \leq x \leq 2\pi.$$

Question 16 (15 marks) Use a SEPARATE writing booklet.

(a) (i) If $I_n = \int_1^e (\log_e x)^n dx$ for $n \geq 0$, show that $I_n = e - nI_{n-1}$ for $n \geq 1$. 3

(ii) Hence, evaluate I_4 . 2

(b) The base of a 3m high painting in the Louvre Museum is 2m above the eye-level of an observer who is x m from the base of the wall on which the painting is hanging. The angle α is the viewing angle at eye-level and is the difference between the angles of elevation of the top and bottom of the painting as seen by the observer.



(i) Show that $\alpha = \tan^{-1}\left(\frac{3x}{x^2 + 10}\right)$. 3

(ii) Hence find how far the observer should stand to maximise the viewing angle. 3

(c) Given that $f(x) = x^6 + 4x^5 - 3x^4 - 8x^3 + 35x^2 - 60x - 225$ has zeroes at $x = \pm\sqrt{5}$ and a double zero, factorise $f(x)$ over:

(i) the real field. 3

(ii) the complex field. 1

Student Number

ASCHAM SCHOOL

YEAR 12 Trial Mathematics Extension 2 Exam

MULTIPLE-CHOICE ANSWER SHEET

- | | | | | |
|-----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1. | A ☐ | B ☐ | C ☐ | D ☐ |
| 2. | A ☐ | B ☐ | C ☐ | D ☐ |
| 3. | A ☐ | B ☐ | C ☐ | D ☐ |
| 4. | A ☐ | B ☐ | C ☐ | D ☐ |
| 5. | A ☐ | B ☐ | C ☐ | D ☐ |
| 6. | A ☐ | B ☐ | C ☐ | D ☐ |
| 7. | A ☐ | B ☐ | C ☐ | D ☐ |
| 8. | A ☐ | B ☐ | C ☐ | D ☐ |
| 9. | A ☐ | B ☐ | C ☐ | D ☐ |
| 10. | A ☐ | B ☐ | C ☐ | D ☐ |

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

SOLUTIONS AND MARKING SCHEME
MATHS EXT 2 HSC TRIAL
EXAM - ASCHAM SCHOOL 2015

Section 1 (Multiple Choice)

$$1. \frac{30}{1-2i} \times \frac{1+2i}{1+2i} = \frac{30(1+2i)}{1+4}$$

$$= \frac{30}{5}(1+2i) = 6+12i$$

C

$$2. (2-i)(2+i) = 4+1 = 5$$

$$5 \times 3 = 15$$

$\therefore$ roots are: $2-i, 2+i, 3$

B

$$3. \frac{b^2}{a^2} = e^2 - 1$$

$$\therefore e^2 = \frac{b^2}{a^2} + 1 = \frac{4}{3} + 1 = \frac{7}{3}$$

$$e = \sqrt{\frac{7}{3}}$$

D

$$4. \text{ if } x = \frac{\alpha+1}{2}$$

$$P(2x-1) = P(2(\frac{\alpha+1}{2})-1)$$

$$= P(\alpha+1-1) = P(\alpha)$$

$$= 0$$

B

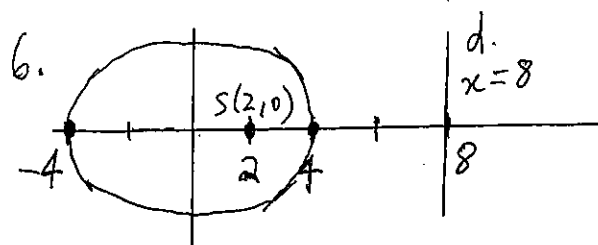
$$5. 3x^2 - y^2 = 11$$

$$\frac{d}{dx}(3x^2 - y^2) = \frac{d}{dx}(11)$$

$$6x - 2y \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{6x}{2y} = \frac{6x-2}{2 \times 1} = -6$$

A



$$a=4 \therefore a^2=16$$

A

7. **C**

$$8. x^2 + y^2 = (x+y)^2 - 2xy$$

$$2 = 1^2 - 2xy$$

$$\therefore xy = -\frac{1}{2}$$

$$x^3 + y^3 = (x+y)(x^2 - xy + y^2)$$

$$= 1(2 - (-\frac{1}{2}))$$

$$= 2\frac{1}{2}$$

C

$$9. 7 - 6x - x^2 = -(x^2 + 6x + 9) + 7 + 9$$

$$= 16 - (x+3)^2$$

$$= 4^2 - (x+3)^2$$

$$\therefore \int \frac{dx}{\sqrt{4^2 - (x+3)^2}} = \sin^{-1}\left(\frac{x+3}{4}\right) + C$$

B

10. A) is true since $x \cdot \cos(x)$ is an odd function
(odd function) $\times$ (even function)

B) is true since e^{-x^2} is an even function.

C) true since $\int_0^\pi \sin 8x dx = 0$

D) False due to greater concavity of x^9 . **D**

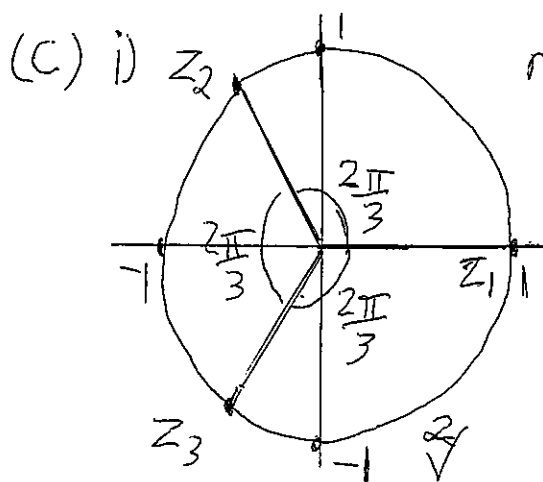
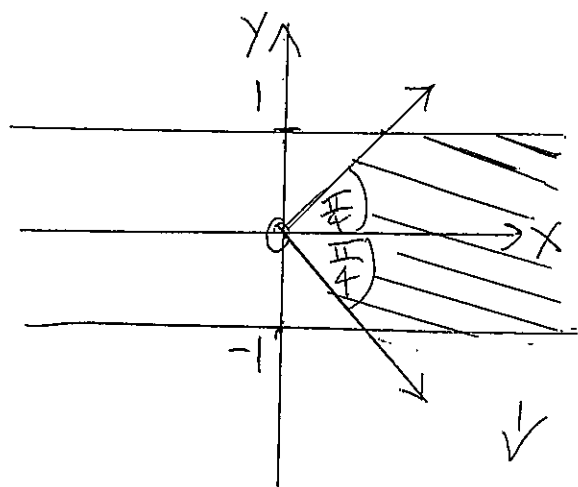
Section 11

Question 11

$$\begin{aligned} \text{(a) i) } z+w &= -2+3i + 3-2i \\ &= 1+i \quad \checkmark \\ &= \sqrt{2} \operatorname{cis}\left(\frac{\pi}{4}\right) \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{ii) } z\bar{w} &= (-2+3i)(3+2i) \quad \checkmark \\ &= -6-4i+9i-6 \\ &= -12+5i \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{(b) let } z &= x+iy \\ \bar{z} &= x-iy \\ z-\bar{z} &= x+iy-(x-iy) \\ &= 2yi \\ |z-\bar{z}| &= |2y| \leq 2 \quad \therefore |y| \leq 1 \quad \checkmark \end{aligned}$$



roots are:

$$z_1 = 1 \operatorname{cis} 0$$

$$z_2 = 1 \operatorname{cis}\left(\frac{2\pi}{3}\right)$$

$$z_3 = 1 \operatorname{cis}\left(\frac{4\pi}{3}\right)$$



$$\text{ii) } z_1 = 1 \operatorname{cis} 0 = 1$$

$$\begin{aligned} z_2 &= 1 \operatorname{cis} \frac{2\pi}{3} = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \\ &= -\frac{1}{2} + i\left(\frac{\sqrt{3}}{2}\right) \end{aligned}$$

$$z_3 = 1 \operatorname{cis} \frac{4\pi}{3} = -\frac{1}{2} - i\left(\frac{\sqrt{3}}{2}\right)$$

$$\begin{aligned} \text{iii) let } w &= z_2 = -\frac{1}{2} + i\left(\frac{\sqrt{3}}{2}\right) \\ \text{then } w^2 &= \left(-\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 i^2 + 2\left(-\frac{1}{2}\right)\left(i\left(\frac{\sqrt{3}}{2}\right)\right) \\ &= \frac{1}{4} - \frac{3}{4} - i\left(\frac{\sqrt{3}}{2}\right) \quad \checkmark \\ &= -\frac{1}{2} - i\left(\frac{\sqrt{3}}{2}\right) = z_3 \end{aligned}$$

$$\begin{aligned} 1+w+w^2 &= 1 + \left(-\frac{1}{2} + i\left(\frac{\sqrt{3}}{2}\right)\right) \\ &\quad + \left(-\frac{1}{2} - i\left(\frac{\sqrt{3}}{2}\right)\right) \quad \checkmark \\ &= 1-1 \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{iv) from iii) } 1+w+w^2 &= 0 \\ \therefore 1+w &= -w^2 \quad \checkmark \\ 1+w^2 &= -w \quad \checkmark \end{aligned}$$

$$\text{also } w^3 = 1 \quad (\text{since } z^3 = 1)$$

$$\begin{aligned} \therefore 1+w^5 &= 1+w^3 \times w^2 \\ &= 1+w^2 \\ &= -w \end{aligned}$$

$$\begin{aligned} 1+w^8 &= 1+w^3 \times w^3 \times w^2 \\ &= 1+w^2 = -w \end{aligned}$$

$$1+w^{11} = 1+(w^3)^3 \times w^2 = -w$$

hence

$$\begin{aligned} (1+w)(1+w^2)(1+w^5)(1+w^8)(1+w^{11}) \\ &= (-w^2)(-w)(-w)(-w)(-w) \\ &= -w^5 = -(w^3)^2 \\ &= -1 \quad \checkmark \end{aligned}$$

Question 12

(a) $\int \cos^4 x \, dx$

$$\begin{aligned}\cos^4 x &= (\cos^2 x)^2 \\ &= \left(\frac{1}{2}(\cos 2x + 1)\right)^2 \\ &= \frac{1}{4}(\cos^2 2x + 2\cos 2x + 1) \\ &= \frac{1}{4}\left(\frac{1}{2}(\cos 4x + 1) + 2\cos 2x + 1\right) \\ &= \frac{1}{8}\cos 4x + \frac{1}{8} + \frac{1}{2}\cos 2x \\ &\quad + \frac{1}{4} \quad \checkmark \\ &= \frac{1}{8}(\cos 4x + 4\cos 2x + 3)\end{aligned}$$

$$\begin{aligned}\int \cos^4 x \, dx &= \frac{1}{8} \int (\cos 4x + 4\cos 2x + 3) \, dx \\ &= \frac{1}{8} \left[\frac{1}{4} \sin 4x + 2\sin 2x + 3x \right] + C \\ &= \frac{1}{32} \sin 4x + \frac{1}{4} \sin 2x + \frac{3}{8}x + C\end{aligned}$$

(b) $\int_0^1 x \cdot e^{-x} \, dx$

Let $\int x \cdot e^{-x} \, dx = \int u \cdot dv$
 where $u = x$ $dv = e^{-x} \, dx$
 $du = dx$ $v = -e^{-x}$ $\checkmark$

$$\begin{aligned}\therefore \int x \cdot e^{-x} \, dx &= uv - \int v \, du \\ &= [-x \cdot e^{-x}] + \int e^{-x} \, dx \quad \checkmark\end{aligned}$$

$$\begin{aligned}\therefore \int_0^1 x \cdot e^{-x} \, dx &= [-x \cdot e^{-x}]_0^1 - [e^{-x}]_0^1 \\ &= [-e^{-1} + 0] - [e^{-1} - e^0] \\ &= 1 - \frac{2}{e} \quad \checkmark\end{aligned}$$

(c) i) $x+7 = A(x+2) + Bx(x+2) + Cx^2$ $\checkmark$
 $= Ax + 2A + Bx^2 + 2Bx + Cx^2$
 $= x^2(B+C) + x(A+2B) + 2A$

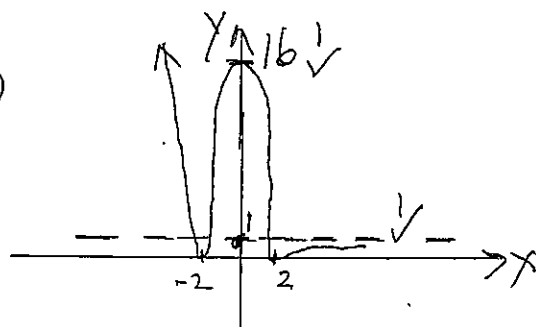
$$\begin{aligned}\therefore 2A &= 7 \quad \text{and} \quad A = \frac{7}{2} \\ A + 2B &= 1 \quad \text{or} \quad \frac{7}{2} + 2B = 1 \\ 2B &= -\frac{5}{2} \\ B &= -\frac{5}{4}\end{aligned}$$

$$\begin{aligned}B + C &= 0 \\ \therefore C &= -B = \frac{5}{4}\end{aligned}$$

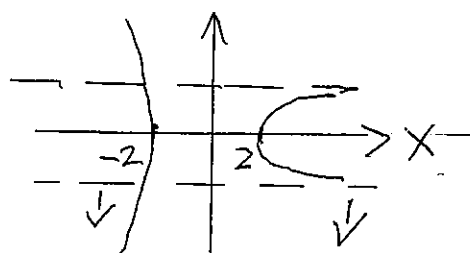
$$\frac{x+7}{x^2(x+2)} = \frac{7}{2x^2} - \frac{5}{4x} + \frac{5}{4(x+2)} \quad \checkmark$$

$$\begin{aligned}\text{ii) } \int \frac{x+7}{x^2(x+2)} \, dx &= \frac{7}{2} \int x^{-2} \, dx - \frac{5}{4} \int \frac{1}{x} \, dx \\ &\quad + \frac{5}{4} \int \frac{1}{x+2} \, dx \quad \checkmark \\ &= -\frac{7}{2x} - \frac{5}{4} \ln x \\ &\quad + \frac{5}{4} \ln(x+2) + C \quad \checkmark\end{aligned}$$

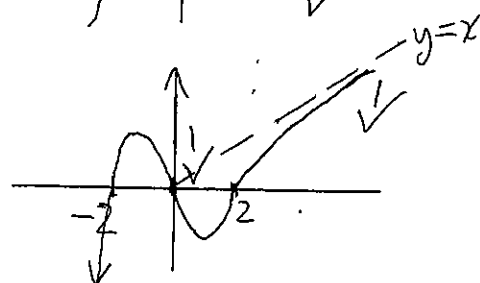
(d) i)



ii)



iii)



Question 13

(a) i) Since ellipse has major axis vertical, it is in the form:

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

$$\therefore b^2 = 9 \text{ and } a^2 = 16$$

$$\frac{b^2}{a^2} = 1 - e^2$$

$$e^2 = 1 - \frac{9}{16}$$

$$= \frac{7}{16}$$

$$e = \frac{\sqrt{7}}{4}$$

coordinates of foci: $(0, \pm ae)$
 $(0, \pm \sqrt{7})$

equations of directrices:

$$y = \pm \frac{a}{e} = \pm 4 \div \frac{\sqrt{7}}{4}$$

$$y = \pm \frac{16}{\sqrt{7}}$$

$$\text{ii) } \frac{d}{dx} \left(\frac{x^2}{9} + \frac{y^2}{16} \right) = \frac{d}{dx} (1)$$

$$\frac{2x}{9} + \frac{2y}{16} \cdot \frac{dy}{dx} = 0$$

$$\frac{y}{8} \frac{dy}{dx} = -\frac{2x}{9}$$

$$\frac{dy}{dx} = -\frac{16x}{9y} = m$$

$$\text{at } P(3 \cos \theta, 4 \sin \theta), m = \frac{-16 \times 3 \cos \theta}{9 \times 4 \sin \theta} = -\frac{4 \cos \theta}{3 \sin \theta}$$

tangent at P:

$$y - y_1 = m(x - x_1)$$

$$y - 4 \sin \theta = -\frac{4 \cos \theta}{3 \sin \theta} (x - 3 \cos \theta)$$

$$(3 \sin \theta) y - 12 \sin^2 \theta = -(4 \cos \theta) x + 12 \cos^2 \theta$$

$$3y \sin \theta + 4x \cos \theta = 12 (\sin^2 \theta + \cos^2 \theta) = 12$$

For C: $y = 4$ in equation of tangent

$$12 \sin \theta + 4x \cos \theta = 12$$

$$4x \cos \theta = 12(1 - \sin \theta)$$

$$x = \frac{12(1 - \sin \theta)}{4 \cos \theta}$$

$$\therefore AC = \frac{3(1 - \sin \theta)}{\cos \theta}$$

For D: $y = -4$ in equation of tangent

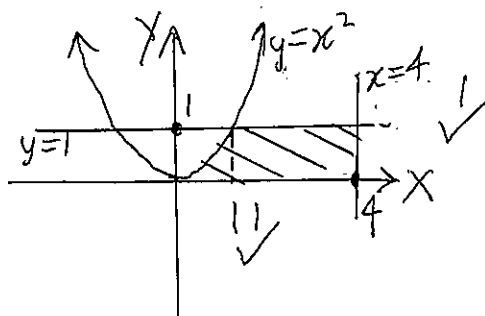
$$-12 \sin \theta + 4x \cos \theta = 12$$

$$x = \frac{12(1 + \sin \theta)}{4 \cos \theta}$$

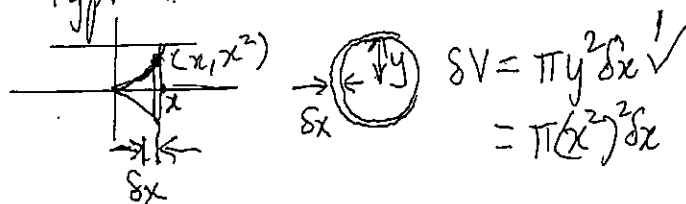
$$\therefore BD = \frac{3(1 + \sin \theta)}{\cos \theta}$$

$$AC \times BD = \frac{3(1 - \sin \theta)}{\cos \theta} \times \frac{3(1 + \sin \theta)}{\cos \theta} = \frac{9(1 - \sin^2 \theta)}{\cos^2 \theta} = 9$$

(b) i)



ii) Split volume into V_1 and V_2 .
 V_1 (for $0 \leq x \leq 1$) and V_2 ($1 \leq x \leq 4$).
 typical slice for V_1 :



$$\delta V = \pi x^4 \delta x$$

$$V_1 = \lim_{\delta x \rightarrow 0} \sum \pi x^4 \delta x = \pi \int_0^1 x^4 dx$$

$$= \pi \left[\frac{x^5}{5} \right]_0^1$$

$$= \frac{\pi}{5} u^3$$

$$V_2 = \pi \int_1^4 12 dx = \pi [x]_1^4 = 4\pi - \pi$$

$$= 3\pi u^3$$

$$V = V_1 + V_2 = \frac{\pi}{5} u^3 + 3\pi u^3 = \frac{16}{5} \pi u^3$$

Question 14

(a) i) $y = c^2 x^{-1}$ $P(c p, \frac{c}{p})$

$$\frac{dy}{dx} = -c^2 x^{-2}$$

$$m_{\text{tangent at } P} = -\frac{c^2}{(\frac{c}{p})^2} = -\frac{1}{p^2}$$

$$\therefore m_{\text{normal}} = p^2 \quad \checkmark$$

$$y - y_P = m_{\text{normal}} (x - x_P)$$

$$y - \frac{c}{p} = p^2 (x - c p) \quad \checkmark$$

$$p y - c = p^3 (x - c p)$$

$$p y - c = p^3 x - c p^4$$

$$p^3 x - p y = c p^4 - c = c(p^4 - 1) \quad (1)$$

ii) sub $Q(c q, \frac{c}{q})$ into (1):

$$p^3 (c q) - p (\frac{c}{q}) = c(p^4 - 1) \quad \checkmark$$

$$c(p^3 q - \frac{p}{q}) = c(p^4 - 1)$$

$$p^3 q^2 - p = p^4 q - q$$

$$p^3 q^2 - p^4 q = p - q$$

$$p^3 q (q - p) = p - q$$

$$p^3 q = \frac{p - q}{-(p - q)} \quad \checkmark$$

$$p^3 q = -1$$

iii) R is MP of PQ

$$\text{i.e. } R(\frac{c p + c q}{2}, \frac{1}{2}(\frac{c}{p} + \frac{c}{q})) \quad \checkmark$$

$$\text{so } x = \frac{c(p+q)}{2} \quad (2) \quad \checkmark$$

$$y = \frac{c(p+q)}{2 p q} \quad (2)$$

the locus of R is

$$c^2(x^2 - y^2)^2 + 4x^3 y^3 = 0 \quad (3)$$

if (1) and (2) satisfy (3).

Sub (1) and (2) into (3):

$$c^2(x^2 - y^2)^2 + 4x^3 y^3$$

$$= c^2 \left(\frac{c^2 (p+q)^2}{4} - \frac{c^2 (p+q)^2}{4 p^2 q^2} \right)^2$$

$$+ 4 \left(\frac{c^3 (p+q)^3}{8} \right) \left(\frac{c^3 (p+q)^3}{8 p^3 q^3} \right) \quad \checkmark$$

$$= c^2 \left(\frac{p^2 q^2 c^2 (p+q)^2 - c (p+q)^2}{4 p^2 q^2} \right)^2$$

$$+ \frac{c^6 (p+q)^6}{16 p^3 q^3}$$

$$= c^2 \left(\frac{c^2 (p+q)^2 (p^2 q^2 - 1)}{16 p^4 q^4} \right)^2 + \frac{c^6 (p+q)^6}{16 p^3 q^3}$$

$$= \frac{c^6 (p+q)^4 (p^2 q^2 - 1)^2 + c^6 (p+q)^6 p q}{16 p^4 q^4}$$

$$= \frac{c^6 (p+q)^4}{16 p^4 q^4} ((p^2 q^2 - 1)^2 + (p+q)^2 p q)$$

Now $(p^2 q^2 - 1)^2 + (p+q)^2 p q$

$$= p^4 q^4 - 2 p^2 q^2 + 1 + (p^2 + 2 p q + q^2) p q$$

$$= p^4 q^4 - 2 p^2 q^2 + 1 + p^3 q + 2 p^2 q^2 + p q^3$$

$$= p^4 q^4 + 1 + (-1) + p q^3$$

$$= p q^3 (p^3 q + 1)$$

$$= p q^3 (-1 + 1) \quad \checkmark$$

$$= 0$$

$\therefore$ (1) and (2) satisfy (3)

and (3) is the locus of R.

Question 14 (continued)

$$(b) i) \ddot{m}x = -m(v+v^3)$$

$$\ddot{x} = -v - v^3 \quad (1)$$

$$v \frac{dv}{dx} = -v - v^3$$

$$\frac{dv}{dx} = -1 - v^2$$

$$\frac{dx}{dv} = -\frac{1}{1+v^2}$$

$$x = -\int \frac{dv}{1+v^2}$$

$$x = -\tan^{-1} v + C \quad \checkmark$$

now $x=0$ when $V=Q$

$$0 = -\tan^{-1} Q + C$$

$$\therefore C = \tan^{-1} Q$$

$$\text{so } x = -\tan^{-1} v + \tan^{-1} Q$$

$$= \tan^{-1} Q - \tan^{-1} v$$

$$\text{Now } \tan x = \tan(\tan^{-1} Q - \tan^{-1} v)$$

$$= \frac{\tan(\tan^{-1} Q) - \tan(\tan^{-1} v)}{1 + \tan(\tan^{-1} Q) \tan(\tan^{-1} v)}$$

$$\tan x = \frac{Q - v}{1 + Qv} \quad \checkmark$$

$$\therefore x = \tan^{-1} \left(\frac{Q-v}{1+Qv} \right) \quad (4)$$

$$ii) \text{ from (1): } \frac{dv}{dt} = -v - v^3$$

$$\therefore \frac{dt}{dv} = \frac{-1}{v+v^3} = -\frac{1}{v(1+v^2)}$$

$$= -\frac{1}{v} + \frac{v}{1+v^2} \quad \checkmark$$

$$\therefore t = -\int \frac{1}{v} dv + \frac{1}{2} \int \frac{2v}{1+v^2} dv$$

$$\therefore t = -\ln v + \frac{1}{2} \ln(1+v^2) + C$$

when $t=0, v=Q$

$$0 = -\ln Q + \frac{1}{2} \ln(1+Q^2) + C$$

$$\therefore C = \ln Q - \frac{1}{2} \ln(1+Q^2)$$

$$\text{and } t = -\ln v + \frac{1}{2} \ln(1+v^2)$$

$$+ \ln Q - \frac{1}{2} \ln(1+Q^2)$$

$$= \ln\left(\frac{Q}{v}\right) + \frac{1}{2} \ln\left(\frac{1+v^2}{1+Q^2}\right)$$

$$= \frac{1}{2} \left(2 \ln\left(\frac{Q}{v}\right) + \ln\left(\frac{1+v^2}{1+Q^2}\right) \right)$$

$$= \frac{1}{2} \left[\ln \frac{Q^2}{v^2} + \ln\left(\frac{1+v^2}{1+Q^2}\right) \right]$$

$$t = \frac{1}{2} \ln\left(\frac{Q^2(1+v^2)}{v^2(1+Q^2)}\right) \quad \checkmark$$

and when $v=R$,

$$t = \frac{1}{2} \ln\left(\frac{Q^2(1+R^2)}{R^2(1+Q^2)}\right) \text{ as required } (2)$$

$$iii) \text{ from (2): } 2t = \ln\left(\frac{Q^2(1+R^2)}{R^2(1+Q^2)}\right) \quad \checkmark$$

$$e^{2t} = \frac{Q^2(1+R^2)}{R^2(1+Q^2)}$$

$$e^{2t} R^2(1+Q^2) = Q^2(1+R^2)$$

$$e^{2t} R^2(1+Q^2) = Q^2 R^2 + Q^2$$

$$e^{2t} R^2(1+Q^2) - Q^2 R^2 = Q^2$$

$$R^2(e^{2t}(1+Q^2) - Q^2) = Q^2$$

$$\therefore R^2 = \frac{Q^2}{e^{2t}(1+Q^2) - Q^2} \quad \checkmark (3)$$

iv) from (3):

$$v^2 = \frac{Q^2}{e^{2t}(1+Q^2) - Q^2}$$

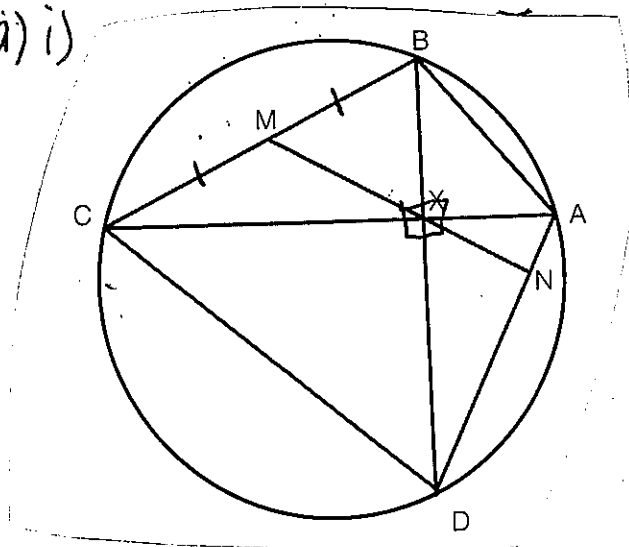
as $t \rightarrow \infty, v^2 \rightarrow 0$ and $v \rightarrow 0 \quad \checkmark$

also from (4):

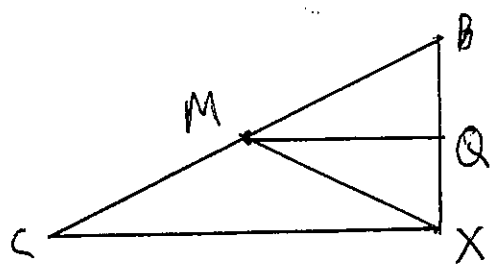
$$x \rightarrow \tan^{-1}(Q) \quad \checkmark$$

Question 15

(a) i)



ii) consider $\triangle BCX$



Construct $MQ \parallel CX$.

$\therefore \angle BQM = \angle BXC = 90^\circ$
(Corresponding $\angle$ s on $\parallel$ lines are $=$)

In $\triangle BMQ$ and $\triangle BCX$;

$$\angle MBQ = \angle CBX$$

$$\angle BQM = \angle BXC = 90^\circ$$

$\therefore \triangle BMQ \parallel \triangle BCX$ (AA) $\checkmark$

$$\frac{BM}{BC} = \frac{BQ}{BX} \quad (\text{common ratio of sides in similar } \triangle s)$$

but $BM = MC$ (given)

$$\therefore \frac{BQ}{BX} = \frac{BM}{BC} = \frac{1}{2}$$

and $BQ = QX$ $\checkmark$

In $\triangle BMQ$ and $\triangle XMQ$;

$$BQ = QX \quad (\text{from above})$$

$$\angle BQM = \angle XQM = 90^\circ$$

MQ is common

$\therefore \triangle BMQ \equiv \triangle XMQ$ (SAS)

$$\therefore \angle MBQ = \angle MXQ$$

(matching $\angle$ s in $\equiv \triangle s$)

$$\text{and } \angle MBX = \angle MXB$$

iii) In $\triangle BMX$;

$MB = MX$ (equal base $\angle$ s of isosceles $\triangle BMX$, $\angle MBX = \angle MXB$)

but $MB = MC$ (given)

$$\therefore MX = MC$$

and $\angle MCX = \angle MXC$ (base $\angle$ s of isosceles $\triangle MCX$)

Now $\angle MXC = \angle AXN$
(vertically opposite $\angle$ s)

$$\therefore \angle MCX = \angle AXN$$

$$\text{or } \angle BCX = \angle AXN \quad (1) \quad \checkmark$$

In the circle $ABCD$;

$$\angle CBD = \angle CAD$$

($\angle$ s on the same chord)

$$\text{i.e. } \angle CBX = \angle XAN \quad (2) \quad \checkmark$$

in $\triangle CBX$ and $\triangle XAN$

$$\angle BCX = \angle XAN \quad (\text{from } (1))$$

$$\angle CBX = \angle XAN \quad (\text{from } (2))$$

$\therefore \triangle CBX \parallel \triangle XAN$ (AA) $\checkmark$

hence $\angle BXC = \angle ANX = 90^\circ$
(matching $\angle$ s in $\parallel \triangle s$)

$\therefore MN \perp AD$

Question 15 (continued)

$$(b) \left(\frac{1}{x}\right)^3 - \left(\frac{1}{x}\right)^2 - 14\left(\frac{1}{x}\right) - 24 = 0 \quad \checkmark$$

$$\frac{1}{x^3} - \frac{1}{x} - \frac{14}{x} - 24 = 0 \quad \checkmark$$

$$\left(\frac{1}{x}\left(\frac{1}{x^2} - 14\right)\right)^2 = \left(24 + \frac{1}{x}\right)^2$$

$$\frac{1}{x}\left(\frac{1}{x^2} - \frac{28}{x} + 14^2\right)x^3 = \left(24^2 + \frac{48}{x} + \frac{1}{x^2}\right)x^3$$

$$1 - 28x + 196x^2 = 576x^3 + 48x^2 + x$$

$$576x^3 - 148x^2 + 29x - 1 = 0 \quad \checkmark$$

$$(c) i) \sin(A+B) = \sin A \cos B + \sin B \cos A$$

$$\sin(A-B) = \sin A \cos B - \sin B \cos A$$

$$\sin(A-B) + \sin(A+B) = 2\sin A \cos B \quad \checkmark$$

$$\text{when } A=2x, B=x$$

$$\sin(2x-x) + \sin(2x+x) = 2\sin 2x \cos x \quad \checkmark$$

$$\therefore \sin x + \sin 3x = 2\sin 2x \cos x \quad \textcircled{1}$$

$$ii) \sin x + \sin 2x + \sin 3x = 0$$

$$2\sin 2x \cos x + \sin 2x = 0 \quad \checkmark$$

(from ①)

$$\sin 2x (2\cos x + 1) = 0$$

$$\sin 2x = 0 \quad \text{or} \quad \cos x = -\frac{1}{2}$$

$$0 \leq 2x \leq 4\pi$$

$$2x = 0, \pi, 2\pi, 3\pi, 4\pi$$

$$\therefore x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$$

$\checkmark$

$$\text{rel } x = \frac{\pi}{3}$$

$$x = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$$

$$= \frac{2\pi}{3}, \frac{4\pi}{3}$$

$\checkmark$

Question 16

$$(a) i) I_n = \int_1^e (\ln x)^n dx$$

$$\text{let } I_n = \int_1^e u dv$$

$$\text{where } u = (\ln x)^n \quad dv = dx$$

$$\frac{du}{dx} = n(\ln x)^{n-1} \times \left(\frac{1}{x}\right) \quad \checkmark \quad v = x$$

$$\int u dv = [uv] - \int v du$$

$$\int_1^e (\ln x)^n dx = \left[x(\ln x)^n \right]_1^e - \int_1^e x \left(\frac{1}{x}\right) n(\ln x)^{n-1} dx$$

$$= \left[e \cdot (\ln e)^n - 1 \cdot (\ln 1)^n \right]$$

$$- n \int_1^e (\ln x)^{n-1} dx \quad \checkmark$$

$$= e \cdot 1^n - n I_{n-1}$$

$$= e - n \cdot I_{n-1}$$

$$ii) I_4 = e - 4I_3$$

$$= e - 4[e - 3I_2]$$

$$= e - 4[e - 3(e - 2I_1)]$$

$$= e - 4[e - 3(e - 2\{e - 1I_0\})] \quad \checkmark$$

$$\text{now } I_0 = \int_1^e (\ln x)^0 dx$$

$$= \int_1^e 1 dx = e - 1$$

$$\therefore I_4 = e - 4[e - 3(e - 2\{e - (e-1)\})]$$

$$= e - 4[e - 3(e - 2 \times 1)]$$

$$= e - 4[e - 3e + 6]$$

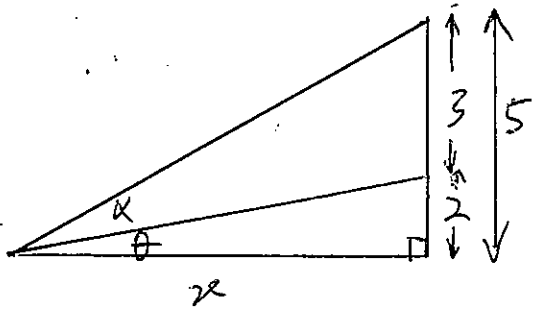
$$= e - 4[6 - 2e]$$

$$= e + 8e - 24$$

$$= 9e - 24 \quad \checkmark$$

Question 1b (continued)

(b) i)



$$\tan(\alpha + \theta) = \frac{\tan \alpha + \tan \theta}{1 - \tan \alpha \tan \theta}$$

$$\frac{5}{x} = \frac{\tan \alpha + \frac{2}{x}}{1 - (\tan \alpha)(\frac{2}{x})}$$

$$\frac{5}{x} \left(1 - \left(\frac{2}{x}\right) \tan \alpha\right) = \tan \alpha + \frac{2}{x}$$

$$\frac{5}{x} - \frac{10}{x^2} \tan \alpha = \tan \alpha + \frac{2}{x}$$

$$\frac{5}{x} - \frac{2}{x} = \tan \alpha + \frac{10}{x^2} \tan \alpha$$

$$\frac{3}{x} = \left(1 + \frac{10}{x^2}\right) \tan \alpha$$

$$\frac{3}{x} = \left(\frac{x^2 + 10}{x^2}\right) \tan \alpha$$

$$\therefore \tan \alpha = \frac{3x^2}{x(x^2 + 10)}$$

$$\alpha = \tan^{-1} \left(\frac{3x}{x^2 + 10} \right)$$

ii) for max α ; $\frac{d\alpha}{dx} = 0$.

$$\frac{d\alpha}{dx} = \frac{1}{1 + \left(\frac{3x}{x^2 + 10}\right)^2} \times \frac{d}{dx} \left(\frac{3x}{x^2 + 10} \right)$$

$$= \frac{3(x^2 + 10) - (3x)(2x)}{(x^2 + 10)^2}$$

$$\frac{(x^2 + 10)^2 + (3x)^2}{(x^2 + 10)^2}$$

$$= \frac{10 - 3x^2}{(x^2 + 10)^2 + (3x)^2}$$

$$= 0 \text{ when } 10 - 3x^2 = 0$$

$$\therefore x = \pm \sqrt{\frac{10}{3}}$$

$$\text{and } x = \sqrt{\frac{10}{3}} \text{ since } x > 0.$$

↓

$x = \sqrt{\frac{10}{3}}$ produces max α since:

$$\frac{d\alpha}{dx} > 0 \text{ when } x < \sqrt{\frac{10}{3}} \text{ and } \frac{d\alpha}{dx} < 0$$

$$\text{when } x > \sqrt{\frac{10}{3}}.$$

↓

c) i) $(x - \sqrt{5})(x + \sqrt{5}) = x^2 - 5$ is a factor of $f(x)$. Hence:

$$\begin{array}{r} x^4 + 4x^3 + 2x^2 + 12x + 45 \\ x^2 - 5 \overline{) } \\ \underline{x^4 + 4x^3 - 5x^2 - 25x - 25} \\ 7x^2 + 37x + 70 \end{array}$$

for double root:

$$\text{let } g(x) = x^4 + 4x^3 + 2x^2 + 12x + 45$$

$$g'(x) = 4x^3 + 12x^2 + 4x + 12$$

$$= x^2(4x + 12) + 4(x + 3)$$

$$g(-3) = (-3)^4 + 4(-3)^3 + 2(-3)^2 + 12(-3) + 45$$

$$= 0 \text{ and } g'(-3) = 0$$

$$\therefore f(x) = (x^2 - 5)(x + 3)^2(x^2 + bx + c)$$

$$(-5) \times (9) \times c = -225 \text{ (equating constant terms)}$$

$$\therefore c = 5$$

$$(-5) \times (6)c + (-5) \times 9 \times b = -60 \text{ (equating terms in } x)$$

$$-30 \times 5 - 45b = -60$$

$$-45b = -60 \implies b = -2$$

$$f(x) = (x - \sqrt{5})(x + \sqrt{5})(x + 3)^2(x^2 - 2x + 5)$$

$$\text{ii) } x^2 - 2x + 5 = x^2 - 2x + 1 + 4 = (x - 1)^2 + (2i)^2$$

$$f(x) = (x - \sqrt{5})(x + \sqrt{5})(x + 3)^2(x - 1 - 2i)(x - 1 + 2i)$$